🌸 Iris Classifier

Project Documentation & User Guide

1. Model Training & Scope

This Machine Learning model is trained exclusively on the famous **Iris Dataset**. The dataset contains historical data for only three specific species of the Iris flower:

- **Iris Setosa**
- **Iris Versicolor**
- **Iris Virginica**

Because the model's knowledge is strictly limited to these three types, it can only predict these species.

2. Limitation with Unknown Data

> **Fundamental ML Rule:** A model can only recognize what it has been trained to see.

If you input measurements for an entirely different flower (such as a Rose or a Sunflower), the model will not be able to identify it as a new flower.

Instead, it will try to match those random numbers with the closest matching patterns of the original three Iris species and force a prediction from one of them.

3. Guide to Testing the App (Exact Values)

To experience how the model successfully changes its predictions based on your inputs, please test the web application using the following standard test scenarios:

Scenario A: Predict "Iris Setosa" (Small Flower)

Sepal Length:	5.0
Sepal Width:	3.4
Petal Length:	1.5
Petal Width:	0.2

Note: *Petal width must be under 0.6 for Setosa.*

Scenario B: Predict "Iris Versicolor" (Medium Flower)

Sepal Length:	6.0
Sepal Width:	2.9
Petal Length:	4.5
Petal Width:	1.3

Note: *Typically ranges between 1.0 and 1.6.*

Scenario C: Predict "Iris Virginica" (Large Flower)

Sepal Length:	6.8
Sepal Width:	3.0
Petal Length:	5.5
Petal Width:	2.1

Note: *Typically stays above 1.8.*